THEME 2 - JewelTech: Agentic Trust from Workshop Scale to Settlement

TOLA

तोला · WEIGH. HEAR. SIGN.

India's Two-Sided Gold Job-Work Ledger

Two Signatures on Every Gram · Zero-Float Deterministic Core · Indic Voice Read-Back

Prepared for

JanAI
MASS ENTREPRENEURSHIP CHALLENGE

TECH4BHARAT
GAVS HACKATHON 2026 · JEWELTECH

THE HIDDEN TRUTH



Every
Jeweller
ISSUES ₹4–5L GOLD
on paper slips and verbal wastage norms (2–3% plain, 20–30% studded)

Every
Karigar
RISKS A MONTH'S PAY
on a single 1-gram dispute (₹13,650–₹15,000 vs ₹15k wage) with zero signed proof

Every
Jewellery ERP
RECORDS ONE SIDE
keeping mutable owner books with no karigar login, voice, or portable work history

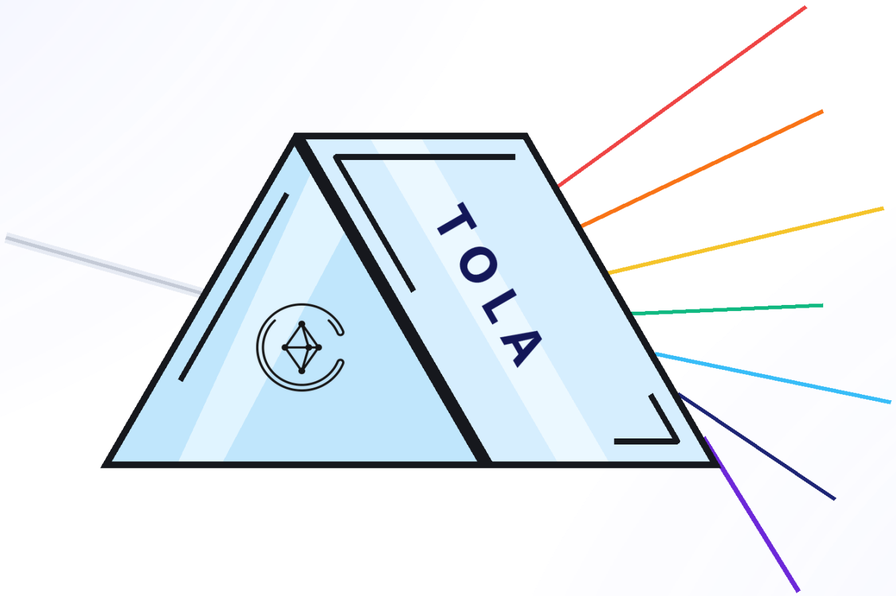
THE MISSING LAYER

PAPER BAHI-KHATA

One-Sided Slip & Verbal Norm
40.00 g Issued · '2.5% Ghaat'
No Karigar Copy · No Signature

WORKSHOP FAILURE MODES

- Unilateral owner-side edits
- Month-end memory disputes
- Unrecorded scrap & dust loss
- Zero collateral for karigar credit



SCALE-LOCKED WEIGHTS
Integer mg · USB/BLE & OCR

VOICE READ-BACK
Sarvam Bulbul v3 in Bengali/Hindi

DUAL SIGNATURES
Jeweller + Karigar Ed25519

HASH-CHAIN LEDGER
Append-only SHA-256 lineage

VARIANCE FLAGS
Deterministic M-05 rule engine

SETTLEMENT & GST
Direct UPI intent + Rule 55 PDF

WORK PASSPORT
Consented 7-day credit proof

GOLD JOB-WORK WEDGE

TODAY

INITIAL CUSTOMER

22K PLAIN GOLD BENCH CLUSTERS (BENGALURU)

- Avenue Road, Chickpet & Raja Market (250+ workshops)
- 5–30 karigars per shop; 15–30s counter handoff limit
- ₹79/karigar/mo entry SaaS (15 yrs < one 1-gram dispute)
- Deployed via JanAI Cafe Doots & KAVMS Jeweller Association

The ideal high-trust wedge into India's ₹4–5L/karigar gold economy.

TOMORROW

Bengaluru Plain Gold Clusters

↓

Kolkata Bowbazar & Singur Hubs

↓

Mumbai Zaveri Bazaar (2.2L Karigars)

↓

Coimbatore, Rajkot & Thrissur

↓

Studded, Kundan & Silver Job-Work

↓

India's 6.1M Artisan Craft Economy

FOUNDATION



CGST Section 143 & Rule 55 Job-Work Challan (PRIMARY)



BIS IS 1417 & HUID 2026 Automated Scale Direction



IIMA & XLRI Gold Wastage Benchmarks (2–3% / 20–30%)



DPDP Act 2023 Spoken Consent & CICRA 2005 Safe Boundary



RFC 8032 Ed25519 + RFC 8785 Canonical JSON Hash Chain

10 mg Scale Capture → Indic Voice Read-Back → Dual Ed25519 Signatures → Weekly Reconcile → Work Passport

MEET TOLA

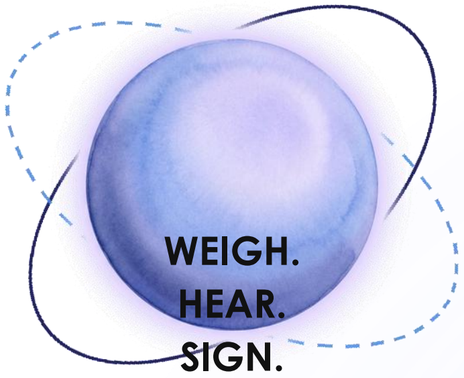
WEIGHS
Captures 10 mg scale reads via USB, BLE, or photo OCR.

SPEAKS
Reads back weight & wastage in Bengali, Hindi, or Kannada.

CO-SIGNS
Locks Jeweller + Karigar Ed25519 keys on every row.

RECONCILES
Flags over-norm variance & records voice rework notes.

Verbal Gold Issue → Two-Sided Cryptographic Truth



One Append-Only Ledger of Workshop Truth

TRANSLATES
Converts colloquial ratti/ana and Bengali voice into JSON.

CHAINS
Links ISSUE, RETURN, SCRAP & DUST in a SHA-256 hash chain.

SETTLES
Triggers direct UPI labour pay & Rule 55 GST challan PDF.

UNLOCKS CREDIT
Exports a consented 7-day Karigar Work Passport.

SYSTEM ARCHITECTURE

BENCH & INPUT SOURCES

- USB-OTG Scale (10 mg)
- BLE GATT Scale (0x181D)
- Scale Display OCR
- Bengali / Hindi Voice
- Scrap & Dust Pouches
- Job-Work Order Norms
- Offline PWA / IVR Queue

Physical Workshop Counter & Feature-Phone Edge Inputs

CAPTURE & VOICE PIPELINE

Scale / OCR Parser → Sarvam Saaras v3 STT → Deterministic Numeral Parser → Spoken-vs-Scale Check → Sarvam Bulbul v3 TTS → Dual Ed25519 Sign Request

Transforms raw scale frames and code-mixed speech into human-confirmed signed events (≤ 20 seconds).

THE BRAIN LEDGER-CORE DAG

- Integer Milligram Math (Zero Float)
- 916 / 750 Fineness Conversion (ppt)
- Wastage Norm Engine (bps)

JobOrder #14 → ISSUE 40.00g → RETURN 38.12g → SCRAP 0.50g

M-05 Flag (+380mg) → Co-Signed Adjust → SHA-256 Settlement

ISSUE → RETURN / SCRAP / DUST → DISPUTE_OPEN → ADJUSTMENT → SETTLEMENT

- M-05 Variance Rules (OVER_NORM)
- RFC 8785 Canonical JSON + SHA-256
- Dual Ed25519 Signature Verifier

LYZR AGENTIC SERVICES

ManagerAgent
Off-hot-path router & prompt-injection guard.

IntakeAgent
Extracts structured slip & rework JSON from speech.

ReconcileAgent
Renders slot-locked neutral bilingual flag summaries.

SettleAgent
Drafts settlement pack & Rule 55 GST challan.

Output Validator
Enforces zero-new-digit & blame-lexicon gates.

GOVERNANCE & TRUST

- ✓ Zero AI Math Invariant
- ✓ Append-Only DB Triggers
- ✓ DPDP Voice Consent Log
- ✓ Tenant RLS Isolation
- ✓ 500-Event CI Eval Gate
- ✓ ₹38/mo AI Cost Cap

ROLE EXPERIENCE & RAILS

- Jeweller Bench Console
- Karigar Voice PWA + IVR
- Lender Work Passport View
- Rule 55 GST Challan PDF

FastAPI 0.115 | PostgreSQL 16 (Append-Only) | Web Serial & Web Bluetooth | NPCI UPI Intent | Cloud Run (asia-south1)

TECHNOLOGY STACK

Indic Voice & Vision AI

Speech-to-Text: Sarvam Saaras v3 (Bengali, Hindi, Kannada)
Text-to-Speech: Sarvam Bulbul v3 (warm read-back)
Fallback STT: Self-hosted IndicConformer 600M
Scale Vision: Gemini 2.5 Flash-Lite 7-segment OCR
Numeral Parser: Deterministic ratti/ana/tola/mg

Governed Agentic Layer

Orchestration: Lyzr Studio (Manager + 3 agents)
IntakeAgent: Slip & rework JSON extraction
ReconcileAgent: Slot-locked bilingual flag cards
SettleAgent: Settlement + Rule 55 challan
Guardrails: Zero-new-digit & blame filter

Backend & Ledger Core

API Runtime: Python 3.12 + FastAPI 0.115
Ledger Engine: Pure-Python integer mg (0 float)
Database: PostgreSQL 16 + append-only triggers
Cache / Queue: Upstash Redis + Cloudflare R2
Isolation: Row-Level Security (tenant_id)

Frontend & Workshop UX

Bench & Phone: Next.js 15 + React 19 + TypeScript
Offline Queue: IndexedDB + Service Worker PWA
Styling & UI: Tailwind CSS 4 + HeroUI / shadcn
Typography: Noto Sans Bengali / Devanagari + Mono
Live Stream: Server-Sent Events (SSE) timeline

Hardware, Telecom & DPI Rails

Wired Scale: Web Serial API (USB-OTG RS-232)
Wireless Scale: Web Bluetooth GATT (0x181D)
Feature Phone: Exotel / Twilio Missed-Call IVR
Labour Payout: NPCI UPI Deep-Link + Setu AA
Compliance: CGST Sec 143 / Rule 55 Challan PDF

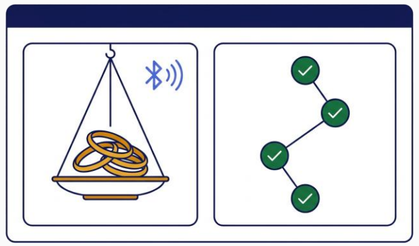
Cryptography, Security & Ops

Signatures: Ed25519 (RFC 8032) via PyNaCl
Hash Chain: RFC 8785 JCS + SHA-256
Device Keys: WebCrypto + Cloud KMS envelope
Deploy & CI: Cloud Run (asia-south1) + GitHub CI
Observability: OpenTelemetry + 500-event eval

USER EXPERIENCE - PROTOTYPE

GITHUB REPO LIVE STAGE DEMO

1. JEWELLER BENCH — SCALE CAPTURE & ISSUE



LIVE SCALE READ (USB / BLE)

40.000 g (40,000 mg)

DECLARED FINENESS & PURE GOLD

22K (916 ppt) → 36,640 mg fine

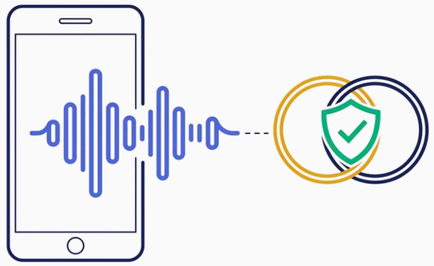
AGREED WASTAGE NORM (2.50%)

250 bps → 1,000 mg allowance

BENCH ACTION

Jeweller signs ISSUE (Ed25519)

2. KARIGAR PHONE — BENGALI VOICE READ-BACK & SIGN



BULBUL v3 VOICE READ-BACK (BENGALI)

“আপনি ৪০ গ্রাম ২২ ক্যারেট সোনা নিচ্ছেন...”

KARIGAR SPOKEN CONFIRMATION

Confirmed on phone / IVR press 1

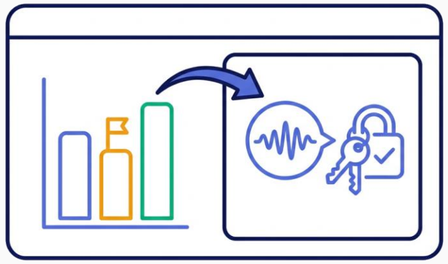
CRYPTOGRAPHIC CO-SIGNATURE

Karigar Ed25519 key locked on row

TOTAL BENCH TIME

18 seconds · Zero typing

3. WEEKLY RECONCILE — +380 MG FLAG & VOICE DISPUTE



RETURN + SCRAP vs EXPECTED

38,120 mg + 500 mg (Loss: 1,380 mg)

DETERMINISTIC M-05 FLAG

+380 mg OVER_NORM_v1 (Neutral)

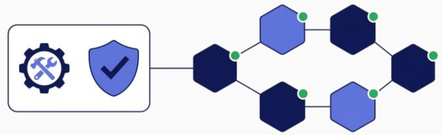
VOICE REWORK EVIDENCE

Solder joint cracked twice (+130 mg)

SETTLEMENT + UPI LABOUR PAY

₹5,040 paid · Rule 55 PDF ready

4. KARIGAR WORK PASSPORT & CHAIN VERIFIER



VERIFIED VOLUME & ORDERS

15 Co-Signed Orders · 194.000 g 22K

DISPUTE-FREE TRACK RECORD

93.3% Clean · +0.18% vs Norm

HASH-CHAIN AUDIT STATUS

8/8 SHA-256 & 16/16 Ed25519 PASS

NBFC / SHG CREDIT EXPORT

7-Day Consented PDF + QR Token

WEIGH. HEAR. SIGN. — 18-Second Bench Capture · Zero Typing · Live in Browser & Feature-Phone IVR

LIVE JOB-WORK SCENARIO

Order #14: Shetty Jewellers (Bengaluru) ↔ Rafiq Sheikh (Karigar) · 4 Plain 22K Bangles

SCALE CAPTURE & RETURN DIFFERENCE

MONDAY ISSUE (CO-SIGNED)

- Gross 22K (916): 40,000 mg
- Fine Gold: 36,640 mg
- Agreed Norm: 2.50% (250 bps)
- Allowed Loss: 1,000 mg
- Signed: Shetty + Rafiq

THURSDAY RETURN (MEASURED)

- Finished Bangles: 38,120 mg
- Katran (Scrap) Return: 500 mg
- Total Accounted: 38,620 mg
- Actual Loss: 1,380 mg (3.45%)
- Over-Norm Delta: +380 mg

Karigar Bengali Voice Note (Saaras v3 STT):
“বাবু, দ্বিতীয় চুড়ির জোড় দুইবার ফেটে গেছে, দুবার রাঙ ঝালাই করতে হয়েছে।”
(“Babu, the second bangle joint cracked twice; had to solder-rework it twice.”)

GOVERNED AGENT FLAG & DUAL RESOLUTION

1. M-05 Variance Engine — FLAGGED

Actual loss 1,380 mg vs 1,000 mg norm → +380 mg (+95 bps).

2. ReconcileAgent Card — OPENED

Slot-locked Bengali + English card; zero blame words.

3. Co-Signed Adjustment — RESOLVED

130 mg solder rework accepted; 250 mg balance settled.

ORDER #14 IMPACT METRICS

40.000 g
22K Issue Co-Signed (36,640 mg Fine Gold)

+380 mg
Over-Norm Variance Caught Deterministically

130 mg
Rework Allowance Co-Signed Without Fight

₹5,040
UPI Labour Settled in 45 Seconds vs 40 Min

GENERATED AUDIT & CREDIT PACK

CGST Rule 55 Job-Work Challan PDF
Auto-filled HSN 7113, 40,000 mg issue, 38,620 mg return & signatures.

8-Event SHA-256 Hash-Chain Receipt
Cryptographic proof: 8/8 hashes valid & 16/16 Ed25519 signatures verified.

Bilingual Settlement Summary + UPI
Bengali & English receipt with UTR reference for ₹5,040 labour payout.

Updated Karigar Work Passport
Order #14 appended: 15 verified orders, 194.000 g volume, QR token.

BHARAT DEPLOYMENT READINESS

Bench-ready. Karigar-ready. Audit-ready.

BENCH & CHANNELS	TRUST & CRYPTO	COMPLIANCE & DPI	UNIT ECONOMICS
USB, BLE & Camera Scale Works with ₹1,500–₹8,000 10 mg workshop scales.	**Dual Ed25519 Co-Sign** Every row requires Jeweller + Karigar keys.	**CGST Sec 143 / Rule 55** One-tap statutory job-work challan PDF.	**₹38 / Month AI Cost** Per active bench across 120 monthly events.
Bengali, Hindi, Kannada Code-mixed ratti/ana/tola & rework speech.	**RFC 8785 + SHA-256 Chain** Tamper-evident hash chain with DB triggers.	**DPDP Act 2023 Consent** Spoken language notice & signed consent hash.	**₹79 / Month Entry Plan** 52% gross margin even on starter tier.
Offline IndexedDB Queue Zero bench downtime during power/net cuts.	**Zero-AI-Math Invariant** 100% deterministic integer milligram arithmetic.	**BIS HUID 2026 Alignment** Ready for automated scale-to-portal rules.	**₹249 / Month Gaddi Pro** 85% gross margin with GST Challan & UPI.
Feature-Phone IVR Fallback Missed-call + Bulbul v3 callback + press 1.	**Neutral Bilingual Flags** Zero blame words; 500-event CI eval gate.	**NBFC Work Passport** Consented 7-day credential outside CICRA.	**JanAI Doot Distribution** Zero-CAC rollout via Bullion & Karigar sanghs.

ROADMAP

1 → 2 → 3 → 4 → 5

1 VALIDATE

DAYS 1–15

Core integer-mg ledger & DAG
Ed25519 + RFC 8785 hash chain
Sarvam Bengali/Hindi voice flow
Lyzr 4-agent governed pipeline
500-event deterministic eval

OUTCOME

Working Dual-Track Prototype (Built Today)

2 STREET PILOT

DAYS 16–30

25 workshops in Raja Market & Chickpet (Bengaluru cluster)
USB-OTG & BLE scale kits live
Weekly Friday hisab on Tola
Karigar Sangha co-sign audits

OUTCOME

25 Benches · 150 Karigars (Days 16–30)

3 CLUSTER LOOP

DAYS 31–60

Expand to Bowbazar (Kolkata) & Zaveri Bazaar (Mumbai)
Feature-phone IVR callback live
Rule 55 GST challan automation
JanAI Doot Sangh distribution

OUTCOME

250 Workshops · 1,500 Karigars (Days 31–60)

4 CREDIT & DPI RAILS

DAYS 61–90

NBFC & Co-op Bank pilots on Karigar Work Passport tokens
NPCI UPI Autopay & Setu AA
Tamil, Gujarati, Marathi packs
Self-hosted IndicConformer

OUTCOME

1,000 Workshops · 1st Loans (Days 61–90)

5 NATIONAL GOLD LEDGER

MONTH 6+

BIS Hallmarking / HUID 2026 automated scale integration
Pan-India cluster coverage
Formal credit for 1M+ karigars
Standard trust layer for gold

OUTCOME

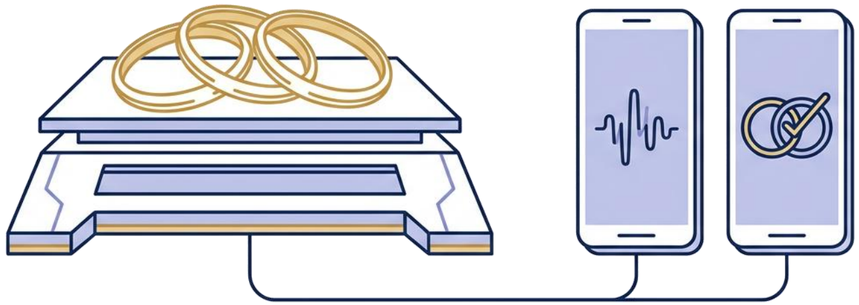
50,000+ Workshops Nationally (Month 6+)

WHY WE ARE READY?

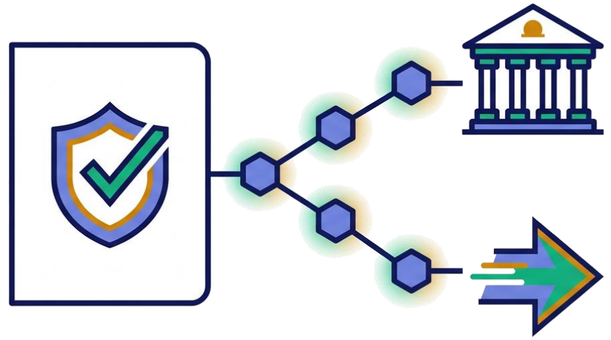
- **Built on Real Goldsmith Workshop Economics & Norms:** Every rule in Tola is anchored in how Indian gaddis actually work — 22K (916) and 18K (750) fineness, item-specific wastage norms (2.5% plain bangles to 5.5% filigree), katran scrap and nyara dust recovery, and CGST Section 143 / Rule 55 job-work challans.

- **Working Dual-Track Codebase & Zero-Float Determinism:** We didn't build a slideware wrapper. Our shared ledger-core enforces integer-milligram conservation, RFC 8785 canonical JSON + SHA-256 hash chaining, dual Ed25519 co-signatures, and a 4-agent Lyzr pipeline where AI never computes or rounds a single milligram.

- **Unit Economics & JanAI Doot Distribution That Scale:** At ₹38/month in AI + telecom cost per active bench against ₹79–₹249/month SaaS tiers, Tola is gross-margin positive from Day 1 — and spreads cluster-by-cluster through Bullion Associations and Bengali Karigar Sanghas at near-zero CAC.



Live Dual-Role Bench + Karigar Voice Prototype (Order #14 Verified)



8/8 SHA-256 Hash-Chain & 16/16 Ed25519 Co-Signature Audit Pass

THANK YOU

TOLA · Two Signatures on Every Gram · Weigh. Hear. Sign.